1. A beam of light consisting of two wavelengths $7000 \mathring{A}$ and $5500 \mathring{A}$ is used to obtain interference pattern in Young's double slit experiment. The distance between the slits is 2.5 mm and the distance between the plane of slits and the screen is 150 cm. The least distance from the central fringe, where the bright fringes due to both the wavelengths coincide, is $n \times 10^{-5}$ m. The value of n is _____.
[2023 (06 Apr Shift 2)]

2. The width of fringe is 2 mm on the screen in a double slit experiment for the light of wavelength of 400 nm. The width of the fringe for the light of wavelength 600 nm will be:
[2023 (08 Apr Shift 2)]
(1) 4 mm
(2) 2 mm
(3) 1.33 mm
(4) 3 mm

3. Unpolarised light of intensity $32 \mathrm{~W~m^{-2}}$ passes through the combination of three polaroids such that the pass axis of the last polaroids is perpendicular to that of the pass axis of first polaroids. If intensity of emerging light is $3 \mathrm{~W~m^{-2}}$, then the angle between pass axis of first two polaroids is _________$^{\circ}$.
[2023 (10 Apr Shift 1)]

4. The ratio of intensities at two points P and Q on the screen in a Young's double slit experiment where phase difference between two waves of same amplitude are $\frac{\pi}{3}$ and $\frac{\pi}{2}$, respectively are
[2023 (10 Apr Shift 2)]
(1) $2 : 3$
(2) $1 : 3$
(3) $3 : 1$
(4) $3 : 2$

5. In a Young's double slit experiment, the ratio of amplitude of light coming from slits is $2 : 1$. The ratio of the maximum to minimum intensity in the interference pattern is
[2023 (13 Apr Shift 2)]
(1) $9 : 4$
(2) $25 : 9$
(3) $2 : 1$
(4) $9 : 1$

6. A single slit of width a is illuminated by a monochromatic light of wavelength 600 nm. The value of a for which first minimum appears at $\theta = 30^{\circ}$ on the screen will be :
[2023 (15 Apr Shift 1)]
(1) $1.2 \mu\mathrm{m}$
(2) $3 \mu\mathrm{m}$
(3) $1.8 \mu\mathrm{m}$
(4) $0.6 \mu\mathrm{m}$

ANSWER KEYS

1. (462) **2.** (4) **3.** (30) **4.** (4) **5.** (4) **6.** (1)

1. (462)

The given data is

$\lambda_1 = 7000\ \overset{\circ}{\mathrm{A}}$

$\lambda_2 = 5500\ \overset{\circ}{\mathrm{A}}$

$d = 2.5 \times 10^{-3}\ \mathrm{m}$

$D = 1.5\ \mathrm{m}$

The path difference is given by

$n\lambda_1 = m\lambda_2$

$7n = 5.5\ \mathrm{m}$

$\Rightarrow 14n = 11\ \mathrm{m} \Rightarrow n = 11$ and $m = 14$

The formula for the distance of a bright fringe is

$y = \frac{n\lambda_1 D}{d}$

$\Rightarrow y = \frac{11\times7\times10^{-7}\times1.5}{2.5\times10^{-3}}$

$= 46.2 \times 10^{-4} = 462 \times 10^{-5}$

It is given that

$n \times 10^{-5} = 462 \times 10^{-5}$

$\Rightarrow n = 462$

2. (4)

For a light of wavelength λ, the fringe width in double slit experiment is given by

$\beta = \frac{\lambda D}{d} \quad \ldots\left(1\right)$

For a light of wavelength λ', the fringe width is given by

$\beta' = \frac{\lambda' D}{d} \quad \ldots(2)$

Divide equation (2) by equation (1) and simplify to obtain the fringe width for the later case.

$\frac{\beta'}{\beta} = \frac{\frac{\lambda' D}{d}}{\frac{\lambda D}{d}}$

$= \frac{\lambda'}{\lambda}$

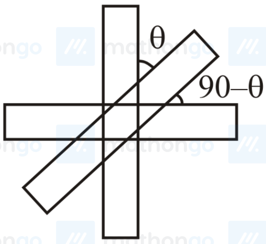
$\Rightarrow \beta' = \frac{\lambda'}{\lambda}\beta \quad \ldots(3)$

Substitute the values of the known parameters into equation (3) to calculate the required fringe width.

$\beta' = \frac{600\ \mathrm{nm}}{400\ \mathrm{nm}} \times 2\ \mathrm{mm}$

$= 3\ \mathrm{mm}$

3. (30)



The given data is
$I_0 = 32\ \mathrm{W\ m^{-2}}$
$I_f = 3\ \mathrm{W\ m^{-2}}$
Thus, by substituting the value, we get
$I_f = \left(\frac{I_0}{2}\cos^2\theta\right)\times\cos^2(90^\circ-\theta)\quad\ldots(i)$
$\Rightarrow 3 = 16\left(\cos^2\theta\cdot\sin^2\theta\right)$
$\Rightarrow \cos\theta\cdot\sin\theta = \frac{\sqrt{3}}{4}$
$\Rightarrow \sin 2\theta = \frac{\sqrt{3}}{2}$
$\Rightarrow \theta = 30^\circ$

4. (4)

The formula to calculate the intensity of the interference pattern in Young's double slit experiment can be written as
$I = 4I_0\cos^2\left(\frac{\phi}{2}\right)\quad\ldots(1)$
Hence, the ratio of the required intensities can be calculated as follows-
$\frac{I_1}{I_2} = \frac{4I_0\cos^2\left(\frac{\pi}{6}\right)}{4I_0\cos^2\left(\frac{\pi}{4}\right)}$
$= \frac{\frac{3}{4}}{\frac{1}{2}}$
$= \frac{3}{2}$

5. (4)

The ratio of the maximum to minimum intensity is given by
$\frac{I_{\max}}{I_{\min}} = \frac{\left(\sqrt{I_1}+\sqrt{I_2}\right)^2}{\left(\sqrt{I_1}-\sqrt{I_2}\right)^2}$
It is given that $\frac{A_1}{A_2} = \frac{2}{1}$
$\frac{I_{max}}{I_{min}} = \left(\frac{A_1+A_2}{A_1-A_2}\right)^2 = \frac{A^2{}_2(2+1)^2}{A^2{}_2(2-1)^2} = \left(\frac{2+1}{2-1}\right)^2 = \frac{9}{1}$

6. (1)

For the first minima
$a\ \sin\theta = \lambda$
The given data is
$\theta = 30^\circ$
$\lambda = 600\ \mathrm{nm}$
The value of a is
$a = \frac{\lambda}{\sin 30^\circ} = \frac{600\times10^{-9}}{\frac{1}{2}} = 1.2\ \mu\mathrm{m}$